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Auditory sensitivity in the fish-catching bat, *Noctilio leporinus*

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Summary. Behavioral and auditory brainstem response (ABR) audiograms are described for the fish-catching bat, *Noctilio leporinus*, which uses short constant frequency/frequency modulated (short-CF/FM) sonar pulses with a CF component of 56–59 kHz, followed by an FM sweep down to 28–32 kHz. Social communication signals contain the frequencies found in sonar pulses, but may extend to lower frequencies (16 kHz).

2. Behavioral thresholds, obtained by operant conditioning in three bats, display maximum sensitivity in the region of the bats' CF sonar component, but show good sensitivity (thresholds less than 10 dB SPL) between 32 and 57 kHz. Below 24 to 32 kHz, sensitivity declined at a rate of 35 to 45 dB/octave. No response was obtained below 1 kHz.

3. The most prominent feature of the behavioral audiogram is an abrupt increase in threshold above 56 to 58 kHz, with an initial roll-off as high as 550 dB/octave. A threshold 'plateau' exists between 64 and 96 kHz, but sensitivity declines rapidly above 96 kHz. No response was obtained above 120 kHz.

4. ABR audiograms were obtained in two animals in which behavioral thresholds had been measured previously. These display a broadly tuned peak of maximum sensitivity at 24 kHz and a more sharply tuned sensitivity peak in the region between 56 and 59 kHz. No responses were obtained below 1 kHz or above 96 kHz.

5. Differences in the shape of the behavioral and ABR threshold curves are discussed. It is suggested that the two sensitivity peaks in the ABR

may result from a disproportionately large representation of frequencies corresponding to social communication signals (24 kHz) and the bat's CF sonar pulse (56 to 59 kHz).

6. The behavioral audiogram of *N. leporinus* is compared to those of other bats, and functional implications are discussed.

Introduction

Members of at least four families of echolocating bats (Emballonuridae, Hipposideridae, Mormoopidae, and Noctilionidae) use a short (less than 5–10 ms) constant frequency (CF) component at the beginning of their sonar signals. While it is thought that most short CF signals are used to improve the echo-detection capabilities of a bat's sonar system (Grinnell 1973; Simmons et al. 1979), some short-CF bats display features which suggest a more specialized use of their CF component. These include: (a) auditory responses in evoked potentials that are sharply tuned to the bat's CF frequency (Grinnell and Hagiwara 1972a; Schuller 1980), and (b) Doppler shift compensation, in which a bat lowers the frequency of its emitted CF signal to compensate for Doppler shifts in echoes (Gustafson and Schnitzler 1979; Schuller 1980).

Sharply tuned auditory responses and Doppler shift compensation are prominent features of the specialized Doppler-sensing sonar of bats that use longer CF echolocation signals (long-CF/FM bats). For instance, the behavioral audiogram of the greater horseshoe bat, *Rhinolophus ferrumequinum*, contains a sensitivity peak which is very sharply tuned to its 83 kHz CF component (Long and Schnitzler 1975). This sharp tuning is thought

Abbreviations: ABR auditory brainstem response; CF constant frequency; FM frequency modulated

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to allow this bat to compensate for the positive Doppler shifts caused by its flight velocity, enabling it to maintain a consistent echo frequency (Schnitzler 1968; Simmons 1974; Schuller et al. 1974). Evidence suggests that the stabilized CF echo provides a carrier for frequency and amplitude modulations caused by insect wingbeats, enabling a bat to detect and identify its prey (Goldman and Henson 1977; Schuller 1979; Pollak and Schuller 1981; Schnitzler and Ostwald 1983). Neural evidence also suggests that the fine frequency resolution of long-CF/FM bats may be employed to measure target velocity (Suga et al. 1983).

In short-CF/FM bats, the function of sharp auditory tuning and Doppler shift compensation has remained unclear. There has not been sufficient study to determine, for example, whether Doppler shift compensation in short-CF/FM bats is always related to sharp auditory tuning, or if Doppler compensation necessarily indicates that CF echoes are processed in a particular manner, e.g., for target detection, for prey identification, or for relative velocity estimates. Furthermore, while auditory evoked potentials have been examined in some of these animals, no behavioral audiograms exist in any short-CF/FM bats.

In this and the following paper, the functional significance of CF signals is examined in the short-CF/FM bat, *Noctilio leporinus*. The foraging behavior of this neotropical animal is unusual even by chiropteran standards. Hunting over fresh and salt water, *N. leporinus* uses its biosonar to detect small water surface disturbances produced by fish and then gaffs the fish with the claws of its enlarged feet (Bloedel 1955; Suthers 1965). The acoustic orientation signals employed by *Noctilio* include a 0–15 ms CF component and an octave-wide FM sweep. These components occur separately under certain circumstances, but are more often combined as a short-CF/FM pulse (Suthers 1965). A CF component of moderate length preceding a very broad FM sweep is relatively unusual in bats (Simmons and Stein 1980), and its significance in relation to the fish-catching bat's foraging behavior is not well understood.

The primary goal of the experiments described in this report was to determine the auditory responsiveness of fish-catching bats to elements of their sonar signals, particularly the extent to which the behavioral audiogram of *Noctilio* is tuned to its short-CF component. A second goal of this study was to obtain evoked potential 'audiograms' from auditory brainstem responses (ABR) and to compare these data with behavioral threshold curves in the same individuals. In many bats, audi-

tory sensitivity has been studied using evoked potential responses from single auditory nuclei, yet these may differ considerably from behavioral audiograms (cf. Grinnell and Hagiwara 1972b; Suthers and Summers 1980). Recent evidence suggests that the ABR may be a better predictor of the shape of behavioral threshold curves (Belknap and Suthers 1982).

The following paper (Wenstrup and Suthers 1984) examines velocity discrimination, Doppler shift discrimination, and Doppler shift compensation in this fish-catching bat.

Materials and methods

One male (NL 7) and two female (NL 9 and NL 21) fish-catching bats (*Noctilio leporinus*), captured in Trinidad, West Indies, were used in these studies. Brief inspections of the outer ears and the echolocation behavior of the subjects revealed no abnormalities. At the time of capture neither female had previously lactated, but both were sexually mature when the testing began. They were probably less than three years of age. The male was sexually mature at the time of capture, its dentition and pelage being indicative of a healthy adult. The lifespan of *N. leporinus*, if similar to other bats, is at least 10–15 years (Tuttle and Stevenson 1982).

Behavioral audiograms

Training and protocol. The behavioral audiogram was measured within an acoustic chamber, the walls and floor of which were lined with convoluted polyurethane foam to reduce echoes. Initially, a bat was placed on a 23 cm by 32 cm horizontal platform covered in the center with cotton cloth and on either side with wire mesh. Each animal was trained to approach a small light at the front of the platform (nearest to the speaker) when this light was turned on. A mealworm reward was given when the bat had placed both forelimbs on the central cloth surface. In the next step, the animal was rewarded when it moved to either side of the platform when the light was turned out. Sound stimuli were then introduced on half of the trials. Each subject learned to associate a reward on the right side of the platform with the presence of the tone burst stimulus, and to associate a reward on the left side with the absence of a tone burst. The area of the central cloth surface, defining the subject's 'listening position', was reduced until it comprised a 10 cm by 3 cm strip just behind the light.

Figure 1 depicts the final arrangement of the platform. When the training was complete, a subject placed its forelimbs on the cotton cloth pad as soon as the light was turned on. The light was turned off 2 s after the animal moved its forelimbs onto the cotton pad. The subject was then allowed 10 s after the light was turned off in which to respond. Movement of both forelimbs off the cloth in one direction constituted a response, for which the animal was either rewarded with a mealworm or mildly punished by turning on the room lights for several seconds before the next trial.

Behavioral thresholds were determined using a variation of the psychophysical method of limits in descending series. Sound stimuli were presented on half of the trials according to a pseudorandom schedule (Gellerman 1933). For trials in which a tone burst was presented, the sound intensity was reduced in 5 dB steps after each correct response until the animal

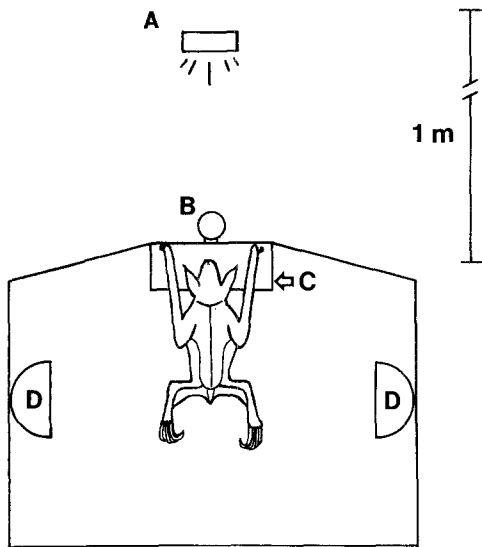


Fig. 1. Response platform for behavioral audiogram. *A* loud-speaker; *B* lightbulb; *C* central cloth surface; *D* reward dishes

made an incorrect response to a tone burst. Threshold for that series was defined as the sound intensity midway between the intensities of the last correct and first incorrect responses to a tone burst. A new series was then begun at an intensity 10 to 15 dB higher than that of the last tone burst to which the bat had correctly responded.

Absolute auditory sensitivity was tested at frequencies between 0.5 kHz and 120 kHz. Since the bats' hearing at frequencies corresponding to their CF pulses was of particular interest, several different frequencies in this region were tested. The sound stimulus consisted of 50 ms pulses with 10 ms rise-fall times and a 5/s repetition rate. For frequencies less than 4 kHz, the duration was increased to 100 ms, with 20 ms rise-fall times, and a 2.5/s repetition rate.

In most cases, behavioral threshold values represent the average of at least eight descending series, collected over two or more days which were often separated by more than a month. If the average threshold for a frequency varied by more than 5 dB on different days, testing was continued until a stable threshold was obtained.

Apparatus. Sound stimuli were generated with an oscillator (Hewlett Packard model 200CD) or function generator (Wave-tek model 136), shaped into pulses by an electronic switch, and then attenuated (Hewlett Packard model 350D attenuator). The frequency of the sine wave generator output was monitored by a frequency counter (Hewlett Packard model 5381A). For frequencies above 40 kHz, the attenuator output was passed through a voltage amplifier, then to a power amplifier (Krohn Hite model DCA 50R), and finally to a condenser speaker (Polaroid model 8665) with a 150 VDC polarization. For frequencies between 4 and 40 kHz, a piezoelectric speaker (Motorola super horn 59067) was substituted for the condenser speaker. For frequencies less than 4 kHz, the attenuator output was fed directly into a Dynaco model 120A power amplifier and an EPI model 100 speaker. At low sound intensities, a fixed attenuator with a value of about 35 dB was inserted into the circuit between the power amplifier and speaker to reduce the amount of internal noise from the power amplifier which reached the speaker. Speakers were located approximately 1 m from the bats' 'listening position'.

The electronic switch which shaped the tone bursts was controlled by a silent switch held by the observer. An identical silent 'dummy' switch was used on trials in which no tone was presented. The observer sat behind the platform, concealed from the bat by a sheet of convoluted polyurethane foam, and controlled the stimulus presentation, light switches, and attenuator settings. The attenuator settings were switched after every trial to prevent the bat from associating the sounds made by turning the attenuator knobs with trials on which a sound stimulus was presented.

The sound pressure level of the acoustic stimulus was measured with a calibrated quarter-inch (6.35 mm) Brüel and Kjaer model 4135 condenser microphone before each day's session. The sound intensity was measured at six positions for calculation of the average sound pressure level at the 'listening position'. The difference in intensity among the six positions was usually less than 3 dB, and always less than 2.5 dB for frequencies between 16 and 64 kHz. The distortion of the system was measured by connecting the output of the calibrated microphone to a Spectral Dynamics model SD 375 signal analyzer, which computed the Fast Fourier Transform (FFT). Second harmonics were attenuated at least 40 dB below the fundamental, while higher harmonics were usually attenuated at least 50 dB.

Evoked potential audiograms

Evoked potential audiograms, obtained from volume-conducted auditory brainstem responses (ABR), were measured in two bats (NL 9 and NL 21) which had previously completed behavioral audiogram testing.

Animal preparation. Surgery was preceded by a 0.2 mg/kg intraperitoneal injection of atropine sulfate (Eli Lilly and Co.). Each animal was anesthetized with intra-muscular injections of ketamine hydrochloride (Ketaset, Bristol Laboratories) and xylazine (Rompun, Haver-Lockhart) at dosages of 20 mg/kg and 1.5–2.0 mg/kg, respectively. A midline incision was made in the skin overlying the skull and a circular tantalum wire electrode (7–9 mm diameter) was inserted just under the skin, centered on the interaural line, and sutured in place. Xylocaine ointment was applied to the open wound. A reference tantalum wire electrode, wrapped in saline-soaked cotton, was placed in the bat's cheek pouch, and a stainless steel ground electrode was inserted into saline-soaked cotton which was wrapped around the animal's foot.

During the experiment, the animal was placed on a heating pad and its rectal temperature was monitored with a thermistor probe. Occasional supplemental doses of ketamine and xylazine were administered at one-third the initial dosage in order to maintain sufficient immobilization and muscle relaxation.

Apparatus. Details of the sound-generating apparatus and acoustic chamber are described in the section concerning the behavioral audiogram. The speakers were placed approximately 1 m from the animal. All surfaces near the bat were covered with cotton or convoluted polyurethane foam to reduce echoes.

Electrical potentials were differentially amplified (Princeton Applied Research, model 113) with a bandpass of 1 Hz to 1000 Hz. The amplifier output was connected to an analog-to-digital conversion input of a computer (Digital Equipment Corp. PDP-8/E) which sampled the input at 4000 Hz. An on-line averaging program computed the average of evoked responses to 512 stimulus presentations and displayed the result on an oscilloscope. Several averaged responses were saved for further analysis.

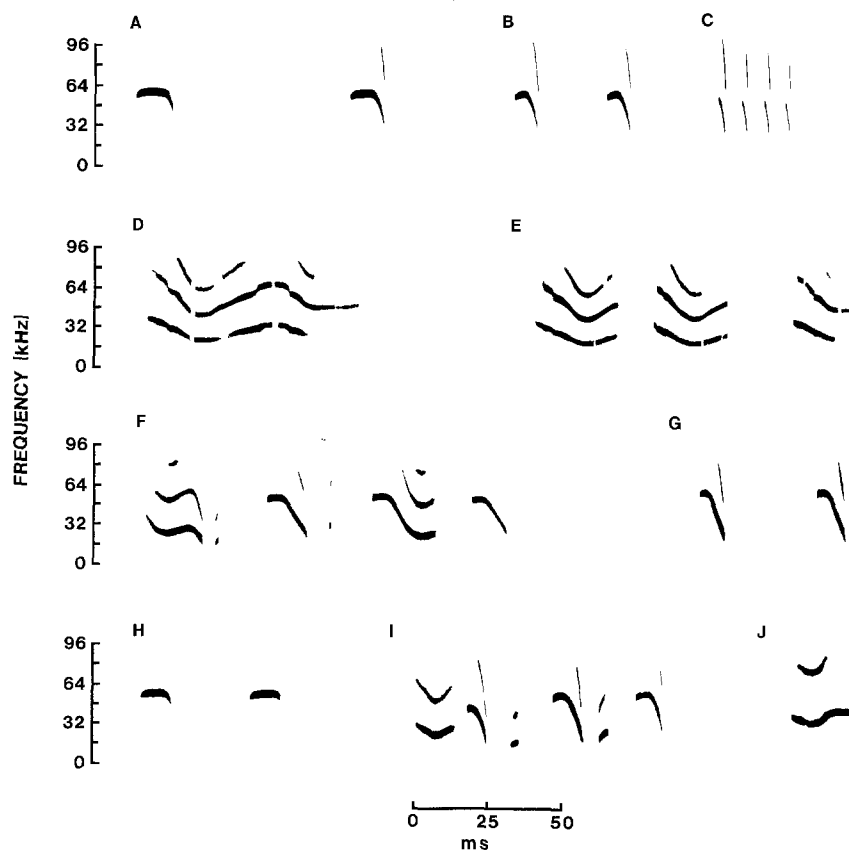


Fig. 2A–J. Sonar signals and common social communication signals of *Noctilio leporinus*. The temporal relationship between each group of pulses is preserved. **A–C** sonar signals produced by resting bat (**A**), and by flying bat during early (**B**) and later (**C**) stages of target pursuit. **D, E** calls of 2-day-old infant when separated from mother. Mother is within hearing range. Its response to these sounds is shown in **F**. **G** 'honks' emitted by one of several bats flying in a large room, used to warn of impending collision. **H–J** vocalizations produced by a male (**H**), female (**I**), or either (**J**) before and during mating

Protocol. Four experiments were performed on each animal. 'Threshold' was defined as the minimum sound intensity capable of evoking a $0.4 \mu\text{V}$ average response. The threshold was determined by reducing the sound intensity in 5 dB steps until the $0.4 \mu\text{V}$ response was no longer obtained. The sound intensity midway between the lowest intensity to elicit the response and the highest intensity that failed to elicit the response was considered to be threshold.

Auditory brainstem responses were measured at frequencies between 0.5 kHz and 120 kHz. The sound stimulus consisted of a 12 ms tone burst with 1 ms rise-fall times, repeated 5 times/s. At sound frequencies less than 10 kHz, the short rise-time of the stimulus spreads energy around the nominal pure tone frequency. However, the power spectra of these low frequency tone bursts, analyzed with the Spectral Dynamics FFT analyzer, were not appreciably widened, and thus the short rise-times probably did not alter significantly the thresholds of ABR responses. Because anesthetic levels fluctuated during the experiments, the order in which different signal frequencies were presented was randomly chosen. The sound pressure level of the acoustic stimulus was measured by placing the calibrated Brüel and Kjaer microphone in the position normally occupied by the bat's head.

Sound recording and analysis

The correspondence between auditory sensitivity curves and vocalizations of fish-catching bats was examined by recording echolocation pulses produced by each of the subjects in these experiments and social communication signals emitted by members of a captive colony of fish-catching bats. The signals were recorded with either QMC (model SM1) or Brüel and

Kjaer condenser microphones, and a Racal model Store 4DS tape recorder. Recorded sonar and social vocalizations were analyzed for frequency-time structure using a spectrum analyzer (Kay model 6061B Sonagraph). The power spectrum of the CF component in sonar sounds emitted by the experimental subjects was measured using the Spectral Dynamics FFT analyzer. This measurement was limited by a tape flutter error of ± 60 Hz. Because the frequency of the CF component changes somewhat within each pulse (usually less than 2 kHz), it should be emphasized that the entire duration of the CF component was included in the spectral analysis. The most intense frequency within this narrow band was termed the 'frequency of the CF'.

Results

Vocalizations

Noctilio employs a CF/FM sonar pulse consisting of a prominent fundamental with a CF component at 56–59 kHz, followed by an FM sweep terminating near 28 kHz (Fig. 2B). Both the CF and FM components occur separately in different orientation situations (Fig. 2A, C). When flying outdoors, these bats usually emit only the fundamental, but add the second harmonic when flying in enclosed spaces (Suthers 1965; Wenstrup and Suthers 1984). For each of the bats in these experiments, the average frequency of the CF in 50 resting sonar pulses is shown by the arrows in Figs. 3–5.

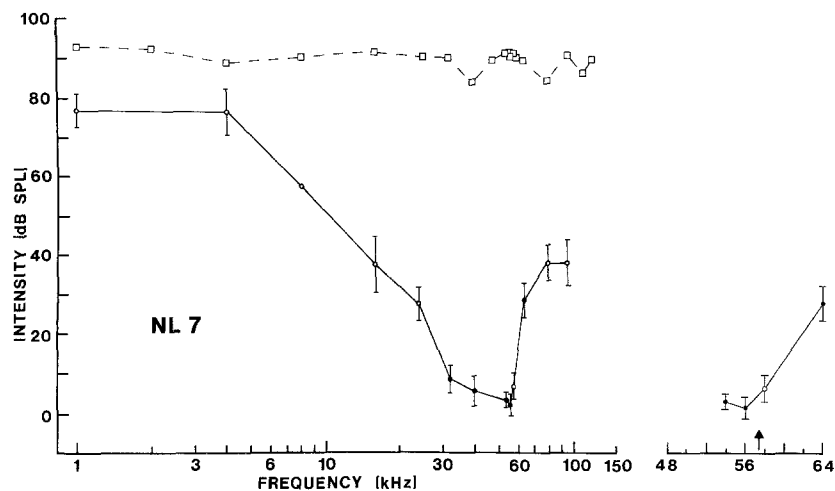


Fig. 3. Behavioral audiogram for one fish-catching bat (*NL 7*), with expansion of the 48 to 64 kHz region at right. Closed circles represent the average of eight or more threshold estimates. Open circles, four to seven threshold estimates. Bars, standard deviations. Broken line at top indicates maximum output of loudspeakers at each tested frequency. Arrow indicates average resting frequency of the CF for 50 sonar pulses recorded from this bat

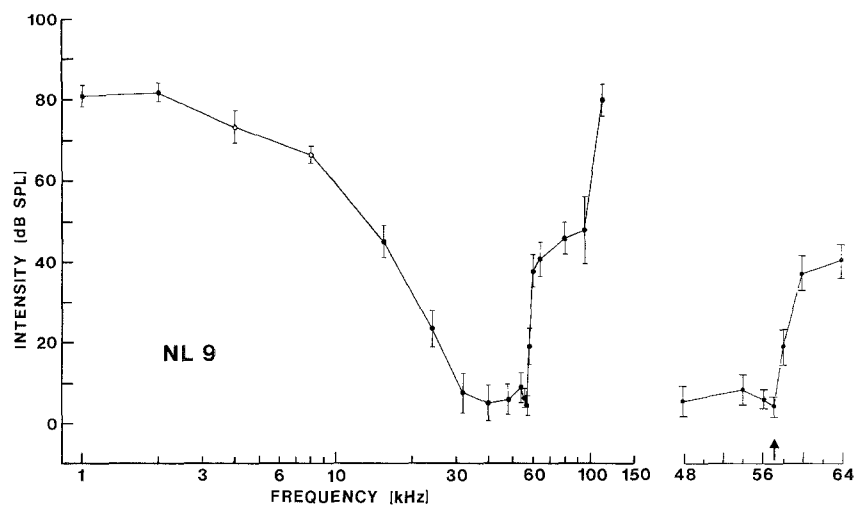


Fig. 4. Behavioral audiogram for *NL 9*, with expansion of the 48 to 64 kHz region at right. All symbols as in Fig. 3

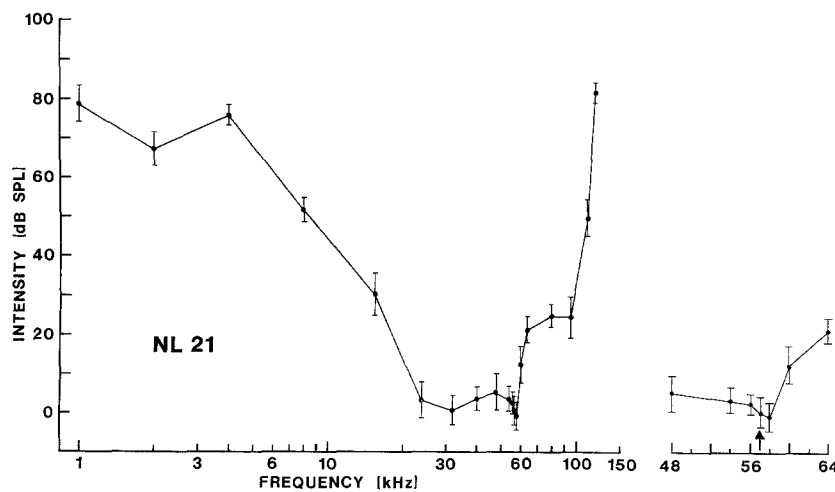


Fig. 5. Behavioral audiogram for *NL 21*, with expansion of the 48 to 64 kHz region at right. All symbols as in Fig. 3

Common social communication sounds used by *N. leporinus* contain most energy between 16 kHz and 64 kHz and usually have two or more harmonics (Fig. 2). The calls of a two day old infant (Fig. 2D, E) also contained CF/FM sounds (not shown) similar to adult sonar signals, but the sonar-like sounds of this infant were lower in frequency (CF near 40 kHz). In adult bats, social vocalizations may contain sonar signals as elements, but the social signals employ additional components, more harmonics, and extend to frequencies as low as 16 kHz (Fig. 2F–J).

Behavioral audiograms

Although individual differences were encountered in the behavioral audiograms of the three bats, there were several common features (Figs. 3–5). Within the range of frequencies corresponding to the fundamental of the bats' orientation signals, sensitivity was very good; thresholds were always less than 10 dB sound pressure level (SPL) between 32 kHz and 57 kHz. Maximum auditory sensitivity occurred near the frequency of the CF (56–58 kHz) in all three bats and ranged from -1 to $+4$ dB SPL. The region of very good sensitivity extended down to 32 kHz in bats NL 7 and NL 9, but NL 21 was still very sensitive at 24 kHz. Below these frequencies, sensitivity initially declined at a rate of 35 to 45 dB/octave. The low frequency limit, expressed as the lowest frequency at which an animal can hear a 60 dB SPL tone burst, averaged 7 kHz in the three bats, but all subjects were able to hear intense sounds as low as 1 kHz. Although the three animals were tested at 0.5 kHz, they did not respond to sound intensities as high as 92 dB SPL.

Immediately above the region of peak sensitivity, there was a very sharp increase in the thresholds of all three animals. The roll-off in this region was about 260 dB/octave for NL 21 and 550 dB/octave for NL 9. NL 7 was not tested at 57 or 60 kHz; this may explain the more gradual roll-off of its sensitivity (150 dB/octave) compared to that found in the other bats. Above this sharp sensitivity decline was a 'plateau' from 64 to 96 kHz, corresponding to a part of the second harmonic of the bats' sonar signals. Sensitivity again decreased rapidly above 96 kHz, and the 60 dB high frequency limit was 101 kHz for NL 9 and 115 kHz for NL 21.

The use of blank (no-stimulus) trials permitted a determination of the overall false alarm rate – the percentage of no-stimulus trials on which the subject moved to its right as if it had heard a

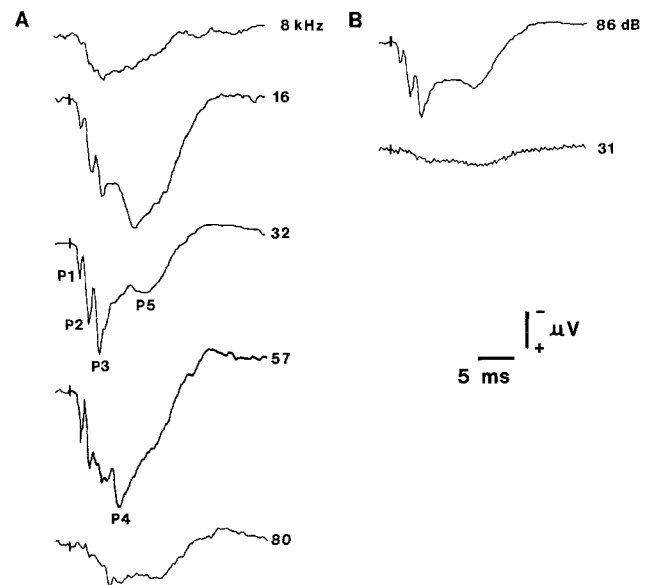


Fig. 6 **A, B.** Auditory brainstem response (ABR) average waveforms showing the effects of stimulus frequency and intensity. Value of vertical calibration bar differs for various traces as specified below. Small vertical marks near the beginning of each waveform indicate time of arrival of the sound stimulus at the bat's ear. **A** ABR averaged waveforms obtained at different frequencies from one experiment on bat NL 21. Stimulus frequency is shown at the right of each trace. P1–P5 are positive peaks in the ABR as discussed in the text. Vertical calibration, 2 μ V, except for 32 kHz trace, where bar equals 4 μ V. **B** Maximum and near-threshold evoked responses to 24 kHz signal in bat NL 21. Sound intensities at right of trace. Threshold at this frequency was 27 dB SPL. Vertical calibration, 86 dB response, 4 μ V; 31 dB response, 1 μ V. Response exceeds threshold at P5 peak

sound. The false alarm rates for NL 7, NL 9, and NL 21 were 14%, 10%, and 9%, respectively.

Evoked potential audiograms

A primary goal of this experiment was to examine the correspondence between ABR thresholds and the behavioral audiogram in *Noctilio*. As discussed later, the ABR provides only a relative measure of auditory responsiveness, one that may not be directly related to auditory sensitivity. It should thus be understood that the terms 'threshold', 'sensitivity', and 'audiogram' have, in the context of the ABR, a specific meaning based on the 0.4 μ V response criterion.

Shape of the ABR. At sound intensities of 80–85 dB SPL, three to five positive peaks (designated P1 through P5) and a slower negative potential appeared in the ABR averaged waveforms (Fig. 6). The average latencies of these positive peaks, independent of frequency, were: P1, 1.4 ± 0.1 ms ($n =$

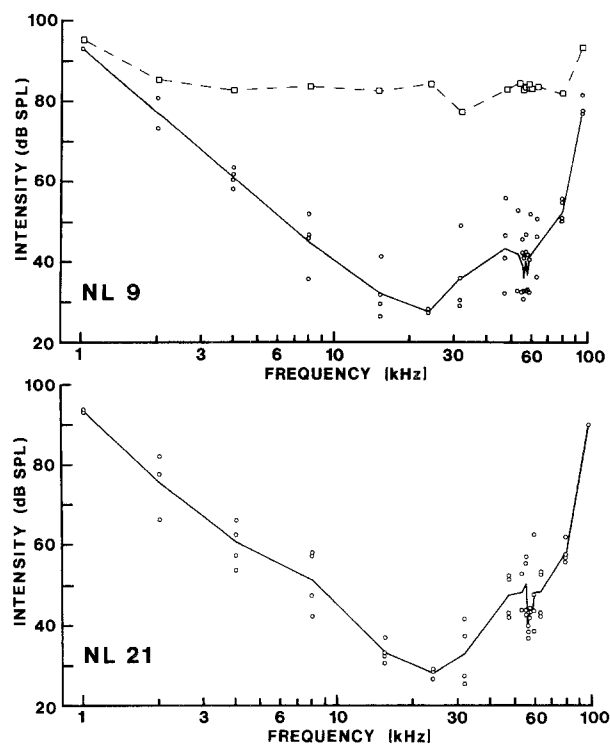


Fig. 7. Threshold curve for ABR in two bats. Open circles display ABR thresholds from four experiments. Solid line connects the average threshold at each tested frequency. Open squares at top of figure display maximum speaker output used for each frequency

10); P2, 2.7 ± 0.2 ms ($n=9$); P3, 4.5 ± 0.5 ms ($n=10$); P4, 6.9 ± 0.2 ms ($n=4$); P5, 11.0 ± 1.2 ms ($n=6$). Decreased stimulus intensity resulted in increased latencies for at least the first three peaks. Any of the positive peaks were used to determine if a response exceeded threshold; in practice P3 and P5 were usually employed, since these peaks persisted at lower sound intensities (Fig. 6B). Only responses within the first 20 ms after the sound reached the bat's ears were used to measure threshold.

Audiograms. Despite the variability in responses obtained at a single frequency in each bat, the evoked potential audiograms of NL 9 and NL 21 were quite similar (Fig. 7). The most sensitive region was between 16 and 32 kHz, with maximum sensitivity occurring at 24 kHz. A second sensitivity peak occurred at the frequency of the CF component, with thresholds 5–10 dB lower than for the surrounding frequencies, but this region generally had thresholds about 10 dB higher than those observed at 24 kHz. No ABR responses were detected at frequencies below 1 kHz (tested at 0.5 and 0.75 kHz) at sound intensities exceeding 90 dB

SPL. Sensitivity above 60 kHz decreased at a rate of about 50 dB/octave. The highest frequency at which the ABR could be observed was 96 kHz, although frequencies of 112 kHz and 120 kHz were tested.

Although auditory evoked responses are reduced at lower brain temperatures in some bats (Harrison 1965), there were no differences between thresholds obtained at rectal temperatures of 37–39 °C and those measured at 32–35 °C. However, when rectal temperature was reduced below 30 °C, both the thresholds and latencies of ABR responses increased.

Discussion

Comparison of auditory threshold curves

The behavioral audiograms of *N. leporinus* (Figs. 3–5) have very low thresholds at the frequencies corresponding to the fundamental sonar component; thresholds increase rapidly outside of this band, except for a sensitivity plateau at the second harmonic of the FM component. The ABR thresholds (Fig. 7) display greatest sensitivity between 16 and 32 kHz, and moderately-sharp tuning at the CF frequency.

The ability to reproduce behavioral thresholds in each bat and the close similarity of the ABR threshold curves of NL 9 and NL 21 suggest that both sets of measurements are reliable. The differences obtained by the two methods probably result from the nature of the two phenomena being measured. The behavioral audiogram relied on a voluntary response when the animal heard a tone burst, presumably reflecting the ability of the auditory nervous system to detect a signal within background noise. Although such measurements are subject to response biases, these threshold curves represent the pure tone sensitivity (i.e. the minimum audible field) of each bat, within the variability of the data indicated in Figs. 3–5.

The ABR potentials represent the synchronized, volume-conducted responses of several neural centers (Jewett 1970). Experiments indicate that there is not a one-to-one correspondence between most of the ABR peaks and the responses of brain-stem auditory centers (Achor and Starr 1980). ABR threshold curves may be influenced not only by auditory sensitivity to tone bursts, but also by the number of neurons responding to a particular frequency, their degree of tuning, the degree of response synchronization, and the threshold criterion. It is a more complex and less easily interpreted measurement.

There are, nevertheless, several advantages to recording extracranial auditory brainstem responses. Such recordings sum the activity of several neural centers, display less frequency bias than do intracranial, microelectrode-recorded evoked potentials, and allow rapid and repeated measurements of auditory responsiveness. These advantages are significant if the shape of the ABR 'audiogram' corresponds to the pure tone sensitivity of an animal. This has been found to be the case for several mammals, including humans (Davis 1976; Koderer et al. 1977), chinchillas (Rothenburg and Davis 1967), and the echolocating megachiropteran bat, *Rousettus* (Belknap and Suthers 1982).

The ABR experiments on *N. leporinus* provided an opportunity to compare behavioral and ABR thresholds on the same individuals in order to determine the success of ABR 'thresholds' in predicting behavioral auditory sensitivity for these bats. Our data show that, at least for *N. leporinus*, the shapes of the behavioral and ABR threshold curves do not correspond closely. This may result from a non-uniform frequency representation within brainstem and midbrain auditory centers of the fish-catching bat. Overrepresentation of the CF component is known to occur in long-CF/FM bats (e.g., Suga and Jen 1976; Schuller and Pollak 1979; Pollak et al. 1983) and also may occur in other bats (Marsh 1981; Jen and Suthers 1982). The ABR thresholds in the fish-catching bat suggest that the 16–32 kHz and 57 kHz regions are overrepresented.

Functional aspects of auditory sensitivity

The abrupt roll-off in sensitivity above the frequency of the CF was the most prominent and unusual feature of the behavioral audiograms of *N. leporinus*. This steep roll-off improves the signal-to-noise ratio of CF sonar echoes by attenuating extraneous high frequency sounds. In conjunction with this aspect of the behavioral curve, moderately sharp tuning to the CF component exists in the ABR. While the interpretation of tuning in the ABR remains speculative, such tuning to the CF in *N. leporinus* may indicate that sharply tuned neurons, a greater neural volume, or both, are devoted to processing this bat's CF echo. Suga et al. (1974) briefly mention that they found no sharp tuning in the N1 response of the fish-catching bat. This suggests that the tuning of the behavioral and ABR responses in this bat is elaborated at stages in the auditory system central to the eighth nerve. If so, this is in contrast to the very sharp tuning

of behavioral and neural responses to the CF echo of horseshoe and mustache bats. In these animals, the sharp tuning to the CF signal found throughout the auditory system appears to be the result of cochlear mechanisms (Bruns 1976; Suga et al. 1975).

Experiments with moving targets indicate that the tuning of *Noctilio's* behavioral audiogram to the frequency of its CF component may also improve this bat's sensitivity to upward frequency (Doppler) shifts by sharply reducing the sensation level of the shifted echo. The sharp roll-off in the audiogram may provide a sensory basis for the Doppler compensation behavior that has been observed in *Noctilio*. Nevertheless, it is clear that the abrupt change in sensitivity near the CF component is not an adaptation for measuring the velocity of slowly moving targets, since this is accomplished with the FM portion of the pulse (Wenstrup and Suthers 1984).

The plateau of sensitivity between 64 and 96 kHz indicates that these bats are sensitive to the second harmonic FM component of their sonar cry. The fish-catching bats in our studies and those of Suthers (1965) routinely employed the second harmonic during the terminal stages of a target pursuit sequence. The reduced amplitude of the second harmonic, combined with the bat's relative insensitivity to its frequency range, suggest that its echo is only audible when target range is less than one to two meters (Wenstrup 1983). The second harmonic is probably of greatest use as a means of extending the bandwidth of short duration, terminal phase pulses to provide more accurate information about target position immediately before intercepting the target.

In general, there is good agreement between the behavioral audiograms and the spectra of social vocalizations used by *N. leporinus* (see Fig. 2). Most social vocalizations include echolocation-type components, but the frequencies employed may extend as low as 16 kHz. Even the lower frequencies of these high intensity signals should be easily heard by nearby bats. The relatively high sensitivity of the ABR responses to frequencies between 16 and 32 kHz suggests that a significant volume of neural tissue may be devoted to processing the lower frequencies of the social communication calls.

Poussin and Simmons (1982) and Ryan et al. (1983) have found that some bats show good sensitivity to low frequency sounds under 2 kHz. There have been several recent reports concerning bats that use frequencies below those in their sonar signals to detect prey by passive listening (e.g. Buchler

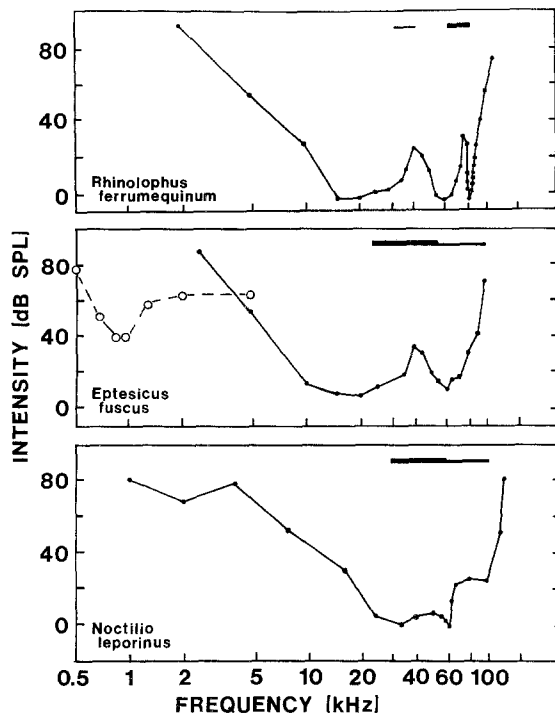


Fig. 8. Behavioral audiograms of three microchiropteran bats using different sonar pulse types. *Rhinolophus ferrumequinum* – long-CF/FM signals; *Eptesicus fuscus* – FM/short-CF signals; *Noctilio leporinus* – short-CF/FM signals. Horizontal bars at top of each audiogram indicate the frequency band of harmonic components in each species' echolocation signal; thickness of bars indicates relative intensity of harmonic components

and Childs 1981; Tuttle and Ryan 1981). We specifically tested low frequency hearing in *Noctilio*, but found no indication of such sensitivity. *N. leporinus* also catches and eats flying insects, but probably does not use low-frequency acoustic cues to detect them.

Fish-catching bats do seem to use passive listening in their feeding behavior, however; they readily attend to splashing sounds (Bloedel 1955; personal observations) and can be coaxed into dipping behavior when such sounds are made. These sounds contain spectral components up to at least 25 kHz. The bats may be able to locate schools of fish swimming at the surface by detecting the splashing sounds they produce.

Behavioral audiograms of other bats

Figure 8 displays the audiograms of *Rhinolophus ferrumequinum* (Long and Schnitzler 1975), *Eptesicus fuscus*, (composite of Dalland 1965, and Pousin and Simmons 1982), and *N. leporinus* (NL 21 of this study). These represent long-CF/FM, FM/short-CF, and short-CF/FM bats, respectively.

Complete behavioral audiograms also exist for *Myotis lucifugus*, an FM bat (Dalland 1965), and *Rousettus aegyptiacus*, an Old World fruit-eating bat that uses click echolocation signals (Suthers and Summers 1980).

One noteworthy difference among the audiograms in Fig. 8 is the extent of tuning at the CF component of the sonar signal, probably reflecting functional differences in the use of CF signals. *Rhinolophus ferrumequinum* has extraordinarily sharp tuning at its CF of 83 kHz, and uses this sharp tuning to compensate for Doppler shifts and to analyze small frequency modulations in echoes. In contrast, *Eptesicus fuscus* displays no sharp threshold changes in the region of its short, 25 kHz CF component. This bat uses FM/short-CF signals in situations requiring the detection of a target (Simmons et al. 1979; Kick 1982), but uses exclusively FM pulses when in laboratory echo-ranging tasks (Simmons 1973) and during the later stages of target pursuit (Simmons et al. 1979). *E. fuscus* does not compensate for Doppler shifts in flight (Jen and Kamada 1982).

The audiogram of *N. leporinus* is less sharply tuned to the CF than that of *Rhinolophus*, and *Noctilio* does not use small echo frequency shifts to obtain Doppler information. Nevertheless, *Noctilio* can obtain Doppler shift information for larger relative velocities, and usually employs Doppler shift compensation when in flight (Wenstrup and Suthers 1984). The broader extent of tuning at the CF reflects the less exacting requirements placed upon its Doppler sensing system compared to *R. ferrumequinum*. With regard to the processing of its CF echoes, *Noctilio* is less specialized than the long-CF/FM bats.

It is also important to realize that *Noctilio* differs significantly from *Eptesicus* regarding certain aspects of its CF signal. While both bats probably use CF signals to improve target detection capabilities, both the audiogram of the fish-catching bat and its Doppler compensation behavior suggest that the processing of such echoes for detection is more sophisticated or that its CF signal is used for other functions as well. The acoustic specializations observed in *Noctilio leporinus* appear to be intermediate between the use of CF signals for detection by FM/short-CF bats (and probably some short-CF/FM bats) and the Doppler sensing strategy of long-CF/FM bats.

Only some bats which use short-CF/FM signals are likely to share the prominent features of the behavioral audiograms obtained from *Noctilio*. These include short-CF/FM hipposiderid bats, in which similar auditory sensitivity curves have been

obtained from evoked potential studies (Grinnell and Hagiwara 1972a; Schuller 1980). Some hipposiderids compensate for Doppler shifts (Gustafson and Schnitzler 1979; Schuller 1980) with greater accuracy than does *Noctilio*, so more sharply tuned audiograms would be expected. In contrast, short-CF/FM emballonurid, mormoopid, and some hipposiderid bats do not display sharply tuned auditory responses (Grinnell 1970; Grinnell and Hagiwara 1972a; Suga et al. 1974; Schuller 1980) and are not known to compensate for Doppler shifts. These bats may use and process CF signals in exactly the same way as FM/short-CF bats.

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